



GOSFORD HIGH SCHOOL

2006

YEAR 12 TRIAL HIGHER SCHOOL CERTIFICATE

MATHEMATICS EXTENSION 2

General Instructions:

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen.
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- All necessary working should be shown in every question.

Total marks: - 120

- Attempt Questions 1 -8
- All questions are of equal value.

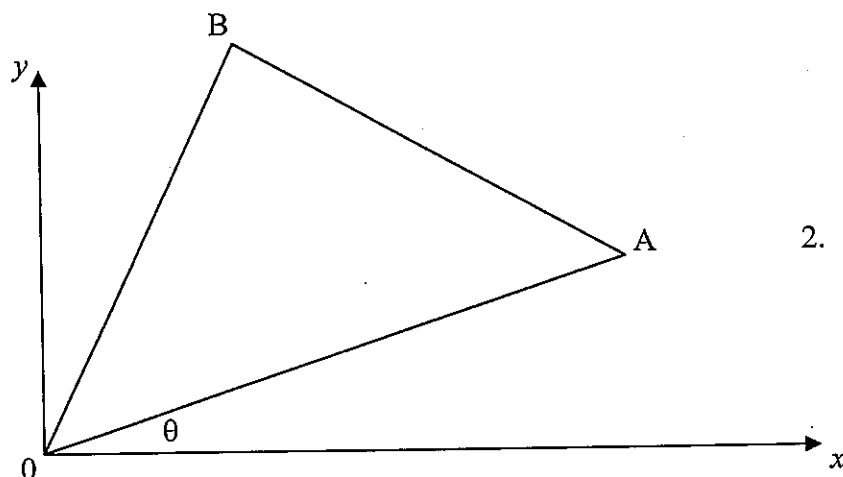
Question 1

- a) Find $\int \frac{dx}{\sqrt{x^2+16}}$. 1.
- b) (i) Find real numbers a and b such that $\frac{x+5}{x^2-2x-3} \equiv \frac{a}{x-3} + \frac{b}{x+1}$ 2.
- (ii) Hence evaluate $\int_4^5 \frac{x+5}{x^2-2x-3} dx$. 2.
- c) Use the substitution $t = \tan \frac{x}{2}$ to evaluate $\int_0^{\frac{\pi}{2}} \frac{dx}{1+\sin x}$. 3.
- d) Use integration by parts to find $\int_0^{\frac{\pi}{4}} x \sec^2 x dx$. 3.
- e) Show that $\int_1^2 (2-x)\sqrt{(x-1)^3} dx = \frac{4}{35}$ 4.

Question 2.

- a) If $Z = \frac{1-7i}{3+4i}$ find:
- i) Z in the form $a+ib$ 1.
- ii) $Z\bar{Z}$ 1.
- iii) $|Z|$ 1.
- iv) $\arg Z$ 1.
- b) Find $\sqrt{-8+6i}$ 2.
- c) Sketch on an argand diagram the locus of the point $P(Z)$ such that $\arg(Z-1) - \arg(Z-i) = \frac{\pi}{4}$. 2.

- d) i) Show that $\cos(\frac{\pi}{3} + \theta) + i \sin(\frac{\pi}{3} + \theta) = \frac{1}{2}(1 + i\sqrt{3})(\cos \theta + i \sin \theta)$ 2.



Let OAB be an equilateral triangle on an Argand diagram where $OA = 1$. The point A represents the complex number Z , where $Z = \text{cis } \theta$.

- ii) Find the complex number represented by B in terms of Z . 2.
- iii) The triangle is now rotated about O through $\frac{\pi}{3}$ radians in an anticlockwise direction to become triangle $OA'B'$. Find the complex numbers represented by the points A' and B' in terms of Z . 3.

Question 3.

- a) $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is an ellipse.
- i) Give the coordinates of S and S' the foci. 1.
- ii) If PSP' is the latus rectum give the coordinates of P . 2.
- iii) Find the equation of the tangent to the ellipse at the point P . 3.
- iv) If the tangent at P meets the minor axis at M prove that the line joining M to the other focus is parallel to the normal at P . 3.
- b) The area enclosed by the curve $y = (x + 3)^2$ and the line $y = 9$ is rotated about the y axis. Using cylindrical shells find the volume formed. 3.

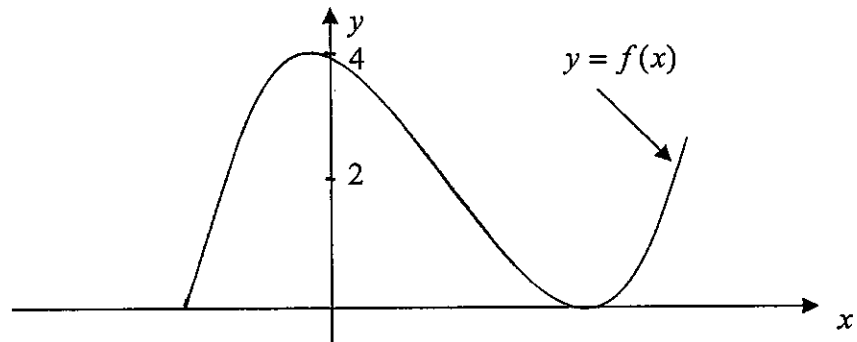
c) Suppose that a lake is stocked initially with 100 fish and that the fish population (P) satisfies hereafter the differential equation

$$\frac{dP}{dt} = k\sqrt{P} \quad (k \text{ a constant}).$$

If after 6 months there are 169 fish in the lake, how many fish will be in the lake after 1 year? 3.

Question 4.

a)



The above graph represents the curve $y = f(x)$. Copy the graph onto your answer sheet and draw an accurate sketch of the graph $y^2 = f(x)$ 2.

b) For the curve $y^2 = x^4(4+x)$

i) Find the x and y intercepts. 2.

ii) By implicit differentiation show that if $y^2 = x^4(4+x)$

then
$$\frac{dy}{dx} = \frac{5x^4 + 16x^3}{2y}$$
 2.

iii) Find any stationary points for the curve. 3.

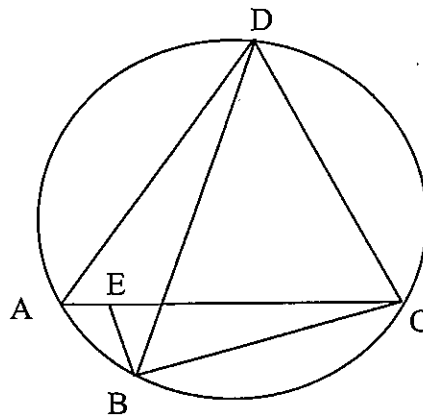
iv) Sketch the curve. 2.

v) By using the substitution $u^2 = 4+x$, or otherwise, find the area of the loop of the curve. 4.

Question 5.

a) The base of a solid is the first quadrant area bounded by the line $4x + 5y = 20$ and the coordinate axes. Find the volume of the solid if every plane section perpendicular to the x axis is a semicircle. 3.

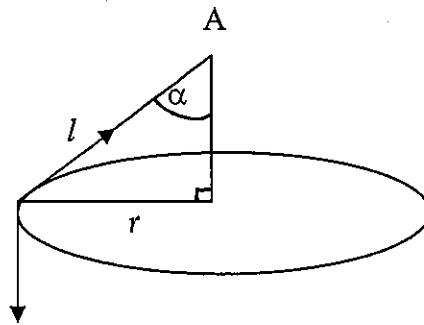
b)



The figure shows a cyclic quadrilateral ABCD with diagonals AC and BD. E is a point on AC such that angle ABE equals angle DBC.

- i) Prove that triangle ABE is similar to triangle DBC 2.
- ii) Prove that triangle ABD is similar to triangle EBC. 2.
- iii) Hence prove Ptolemy's theorem, which states
 $BA \times DC + AD \times BC = AC \times BD$ 3.

c)



- i) A particle of mass m that is connected to a light string of length l to a fixed point A , describes with uniform speed a horizontal circle, radius r , whose centre is vertically below A . If the semi-vertical angle is α show that the tension (T) in the string is given by $T = \frac{mg}{\cos \alpha}$ and that the linear velocity (v) is given by 3.

$$v = \sqrt{rg \tan \alpha}, \text{ where } g \text{ is the force due to gravity.}$$

- ii) A string 60 cm long can sustain a mass of 10 kg vertically hung on it. Find the maximum semi-vertical angle and the number of revolutions per minute a particle of 5 kg mass can make before the string breaks. 2.

Question 6.

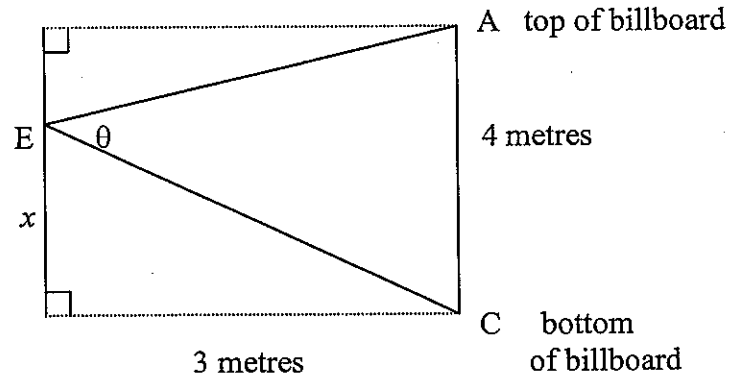
a) $P(x)$ is a polynomial of degree 4 with real coefficients.

- i) Show that if the complex number α is one zero of $P(x)$ then its conjugate $\bar{\alpha}$ is also a zero of $P(x)$. 2.

- ii) The complex number α satisfies $\text{Im}(\alpha) \neq 0, \text{Re}(\alpha) = a$ and $|\alpha| = r$. Show that if α is a zero of $P(x)$, then $P(x)$ has a factor $x^2 - 2ax + r^2$ over R , the field of real numbers. 2.

- iii) α is a non-real double zero of $P(x) = x^4 - 8x^3 + 30x^2 - 56x + 49$. Factor $P(x)$ into irreducible factors over R and find the four roots of $x^4 - 8x^3 + 30x^2 - 56x + 49 = 0$ 4.

b)



- i) Given $\sin(\alpha + \beta) = \sin \alpha \cos \alpha + \cos \alpha \sin \beta$ and $\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta$
 prove $\tan(\alpha + \beta) = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta}$ 2.

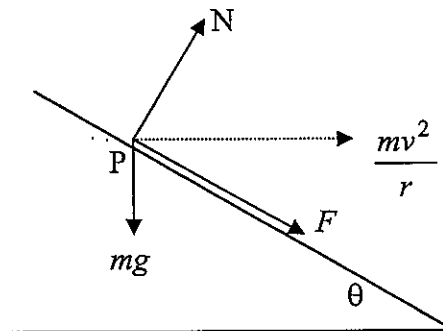
- ii) A billboard AC, 4 metres high is to be positioned vertically and parallel to a highway and at a height which maximises $\tan \theta$, where θ is the angle subtended at E. E represents the eyes of passengers on the top deck of passing double decker buses. The billboard must be 3 metres from the passengers. Show that : 2.

$$\tan \theta = \frac{12}{9 - 4x + x^2} \quad \text{where } x \text{ is the distance DE.}$$

- iii) Find how far below the eyelevel of passengers the base of the billboard must be for θ to be a maximum. 3.

Question 7.

a)



The diagram shows the forces acting on a point P which is moving on a banked circular track. The point P has a mass m and is moving in a horizontal circle of radius r with uniform speed v . The track is inclined at an angle θ to the horizontal. The point experiences a normal reaction force N from the track, a vertical force of magnitude mg due to gravity and a force due to friction F so that the net force on the particle is a force of magnitude $\frac{mv^2}{r}$ directed towards the centre of the horizontal circle. By resolving the forces at P into their horizontal and vertical components show that:

$$F = \frac{mv^2}{r} \cos \theta - mg \sin \theta \quad 3.$$

b) i) Show that the normal to the hyperbola $xy = c^2$ at the point $P(ct, \frac{c}{t})$

$$\text{has the equation } y = t^2 x + \frac{c}{t} - ct^3. \quad 2.$$

ii) If the normal at P meets the line $y = x$ at N , and the tangent at P meets $y = x$ at T , find the coordinates of N and T . 2.

iii) If O is the origin, prove that $OT \cdot ON = 4c^2$ 3.

c) i) Divide x^3 by $(x+1)$ 2.

$$\text{ii) Find } \int \frac{dx}{x^{\frac{1}{2}} + x^{\frac{1}{3}}} \quad 3.$$

Question 8.

a) Show $\ddot{x} = v \frac{dv}{dx}$ 2.

- b) i) A particle of mass m is projected vertically upwards under gravity (g), the air resistance to the motion being $\frac{mgv^2}{a^2}$ when the speed is v , where a is a constant.

Show that during the upward motion of the particle

$$v \frac{dv}{dx} = -\frac{g}{a^2}(a^2 + v^2) \quad 2.$$

where x is the upward vertical displacement.

- ii) show that the greatest height reached, given the speed

of projection u , is: $\frac{a^2}{2g} \ln(1 + \frac{u^2}{a^2})$. 3.

- c) A polynomial of degree n is given by $P(x) = x^n + ax - b$. It is given that the polynomial has a double root at $x = \alpha$.

i) Find the derived polynomial $P'(x)$ and show that $\alpha^{n-1} = -\frac{a}{n}$. 2.

ii) Show that $\left(\frac{a}{n}\right)^n + \left(\frac{b}{n-1}\right)^{n-1} = 0$ 4.

iii) Hence deduce that the double root is $\frac{bn}{a(n-1)}$ 2.

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, x \neq 0 \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, a > 0, -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln(x + \sqrt{x^2 - a^2}), x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln(x + \sqrt{x^2 + a^2})$$

NOTE:

$$\ln x = \log_e x, x > 0$$

2006 Exat2. Trial: SOLUTIONS

Ques 1.

$$a) \int \frac{dx}{\sqrt{x^2+16}}$$

$$= \ln(x + \sqrt{x^2+16})$$

(from standard integrals)

$$b) \frac{x+5}{x^2-2x-3} = \frac{a}{x-3} + \frac{b}{x+1}$$

$$(i) \frac{x+5}{(x-3)(x+1)} = \frac{a}{x-3} + \frac{b}{x+1}$$

$$x+5 = a(x+1) + b(x-3)$$

$$\text{let } x=3: 8 = 4a$$

$$2 = a$$

$$\text{let } x=-1: 4 = -4b$$

$$-1 = b$$

$$\therefore \frac{x+5}{x^2-2x-3} = \frac{2}{x-3} - \frac{1}{x+1}$$

$$ii) \int_4^5 \frac{x+5}{x^2-2x-3} dx$$

$$= \int_4^5 \left(\frac{2}{x-3} - \frac{1}{x+1} \right) dx$$

$$= \left[2\ln(x-3) - \ln(x+1) \right]_4^5$$

$$= \left[\ln \left(\frac{(x-3)^2}{x+1} \right) \right]_4^5$$

$$= \ln \frac{16}{6} - \ln \frac{1}{5}$$

$$= \ln \left(\frac{10}{3} \right)$$

$$c) \int_0^{\pi/2} \frac{dx}{1+\sin x}$$

$$t = \tan \frac{x}{2}$$

$$x = \pi/2: t = 1; x = 0, t = 0$$

$$= \int_0^1 \frac{1}{1+\frac{2t}{1+t^2}} \cdot \frac{2}{1+t^2} dt$$

$$= \int_0^1 \frac{2}{t^2+2t+1} dt$$

$$= \int_0^1 \frac{2}{(t+1)^2} dt$$

$$= \left[\frac{-2}{t+1} \right]_0^1$$

$$= -1 - (-2)$$

$$= 1$$

$$d) \int_0^{\pi/4} x \cdot \sec^2 x dx$$

$$= \int_0^{\pi/4} x \cdot \frac{d}{dx} (\tan x) dx$$

$$= \left[x \tan x \right]_0^{\pi/4} - \int_0^{\pi/4} \tan x dx$$

$$= \pi/4 - \int_0^{\pi/4} \frac{\sin x}{\cos x} dx$$

$$= \pi/4 + \left[\ln(\cos x) \right]_0^{\pi/4}$$

$$= \pi/4 + \ln(\frac{1}{\sqrt{2}}) - \ln 1$$

$$= \pi/4 - \frac{1}{2} \ln 2$$

$$e) \int_1^2 (2-x) \sqrt{(x-1)^3} dx$$

$$= \int_1^2 2(x-1)^{3/2} dx - \int_1^2 x(x-1)^{3/2} dx$$

$$= \left[\frac{4(x-1)^{5/2}}{5} \right]_1^2 \quad \begin{array}{l} \text{let } u = x-1 \\ \frac{du}{dx} = 1 \\ du = dx \end{array}$$

$$x=1, u=0$$

$$x=2, u=1$$

$$= \frac{4}{5} - \int_0^1 (u+1)u^{3/2} du$$

$$= \frac{4}{5} - \int_0^1 u^{5/2} + u^{3/2} du$$

$$= \frac{4}{5} - \left[\frac{2}{7} u^{7/2} + \frac{2}{5} u^{5/2} \right]_0^1$$

$$= \frac{4}{5} - \left(\left(\frac{2}{7} + \frac{2}{5} \right) - 0 \right)$$

$$= \frac{4}{35}$$

Ques 2

a) i) $\frac{1-7i}{3+4i}$

$$= \frac{1-7i}{3+4i} \times \frac{3-4i}{3-4i}$$

$$= \frac{3-4i-21i-28}{9+16}$$

$$= \frac{-25-25i}{25}$$

$$= -1-i$$

ii) $z \bar{z}$

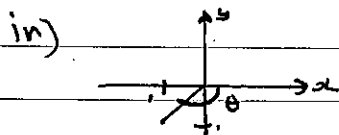
$$= (-1-i)(-1+i)$$

$$= 2$$

iii) $|z|$

$$= \sqrt{1^2+1^2}$$

$$= \sqrt{2}$$



$$\arg z = -\frac{3\pi}{4}$$

b) $\sqrt{-8+6i}$

let $a+ib = \sqrt{-8+6i}$

$$a^2 - b^2 + 2iab = -8 + 6i$$

$$\therefore a^2 - b^2 = -8 \quad \dots (1)$$

$$2ab = 6$$

$$ab = 3 \quad \dots (2)$$

from (2) $b = \frac{3}{a}$; sub into (1)

$$a^2 - \frac{9}{a^2} = -8$$

$$a^4 - 9 = -8a^2$$

$$a^4 + 8a^2 - 9 = 0$$

$$(a^2+9)(a^2-1) = 0$$

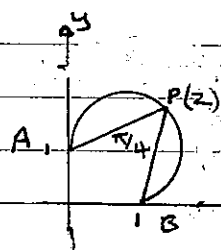
$$a^2 = -9 \quad \text{or} \quad a^2 = 1$$

no solⁿ. $a = \pm 1$

$$\therefore b = \pm 3$$

$$\therefore \sqrt{-8+6i} = 1+3i \quad \text{or} \quad -1-3i$$

c)



Locus is arc APB

d) i) $\cos(\frac{\pi}{3}+\theta) + i\sin(\frac{\pi}{3}+\theta) = \frac{1}{2}(1+i\sqrt{3})(\cos\theta + i\sin\theta)$

L.H.S

$$= \cos\frac{\pi}{3}\cos\theta - \sin\frac{\pi}{3}\sin\theta + i\sin\frac{\pi}{3}\cos\theta + i\cos\frac{\pi}{3}\sin\theta$$

$$= \frac{1}{2}\cos\theta - \frac{\sqrt{3}}{2}\sin\theta + \frac{i\sqrt{3}}{2}\cos\theta + \frac{i}{2}\sin\theta$$

$$= \frac{1}{2}(\cos\theta - \sqrt{3}\sin\theta + i\sqrt{3}\cos\theta + i\sin\theta)$$

$$= \frac{1}{2}(\cos\theta + i\sin\theta + \sqrt{3}(i\cos\theta - \sin\theta))$$

$$= \frac{1}{2}(\cos\theta + i\sin\theta + i\sqrt{3}(\cos\theta + i\sin\theta))$$

$$= \frac{1}{2}(1+i\sqrt{3})(\cos\theta + i\sin\theta)$$

$$= \text{R.H.S}$$

let A be the point $\cos\theta + i\sin\theta \equiv Z$

ii) For B. mod. = 1 and

$$\arg = \theta + \frac{\pi}{3} \quad (\triangle OAB \text{ isosceles})$$

$$\therefore B \text{ is the point } \cos(\theta + \frac{\pi}{3}) + i\sin(\theta + \frac{\pi}{3})$$

$$= \frac{1}{2}(1+i\sqrt{3})(\cos\theta + i\sin\theta)$$

$$= \frac{1}{2}(1+i\sqrt{3})Z$$

3.

iii) When ΔOAB is rotated through $\pi/3$ A' will coincide with B .

Hence $A' = B = (1+i\sqrt{3})Z$

Now B' will have a modulus of 1 and an argument of $2\pi/3$.

$$\therefore B' = \cos(2\pi/3 + \theta) + i\sin(2\pi/3 + \theta)$$

$$= -\frac{1}{2}\cos\theta - \frac{\sqrt{3}}{2}\sin\theta + i\left(\frac{\sqrt{3}}{2}\cos\theta - \frac{1}{2}\sin\theta\right)$$

$$= -\frac{1}{2}\cos\theta - \frac{\sqrt{3}}{2}\sin\theta + \frac{i\sqrt{3}}{2}\cos\theta - \frac{i}{2}\sin\theta$$

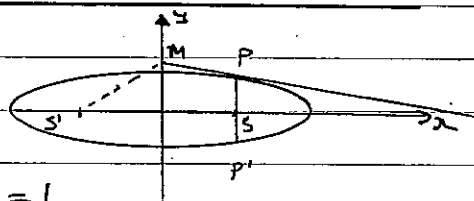
$$= -\frac{1}{2}(\cos\theta + i\sin\theta) + \frac{i\sqrt{3}}{2}(\cos\theta + i\sin\theta)$$

$$= \frac{1}{2}(i\sqrt{3} - 1)(\cos\theta + i\sin\theta)$$

$$= \frac{1}{2}(i\sqrt{3} - 1)Z$$

Ques. 3

a)



$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

i) $S(ae, 0)$, $S'(-ae, 0)$

ii) $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ -----(1)

$x = ae$ -----(2)

Sub (2) into (1)

$$\frac{a^2 e^2}{a^2} + \frac{y^2}{b^2} = 1$$

$$e^2 + \frac{y^2}{b^2} = 1$$

$$\frac{y^2}{b^2} = 1 - e^2$$

$$= 1 - \left(1 - \frac{b^2}{a^2}\right)$$

$$\frac{y^2}{b^2} = \frac{b^2}{a^2}$$

$$y^2 = \frac{b^4}{a^2}$$

$$y = \pm \frac{b^2}{a}$$

$\therefore P(ae, \frac{b^2}{a})$

iii) $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

$$\frac{2x}{a^2} + \frac{2y}{b^2} \cdot \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = -\frac{b^2 x}{a^2 y}$$

at $(ae, \frac{b^2}{a})$: $\frac{dy}{dx} = \frac{-b^2 \cdot ae}{a^2 \cdot \frac{b^2}{a}} = -e$

$\therefore$ equation of the tangent

$$y - \frac{b^2}{a} = -e(x - ae)$$

$$y - \frac{b^2}{a} = -ex + ae^2$$

$$y = -ex + ae^2 + \frac{b^2}{a}$$

iv) for M $x = 0$

$$y = ae^2 + \frac{b^2}{a}$$

$\therefore M(0, ae^2 + \frac{b^2}{a})$

gradient $MS' = \frac{ae^2 + \frac{b^2}{a} - 0}{0 - (-ae)}$

$$= \frac{ae^2 + \frac{b^2}{a}}{ae}$$

$$= \frac{a\left(1 - \frac{b^2}{a^2}\right) + \frac{b^2}{a}}{ae}$$

$$= \frac{a - \frac{b^2}{a} + \frac{b^2}{a}}{ae}$$

$$= \frac{a}{ae}$$

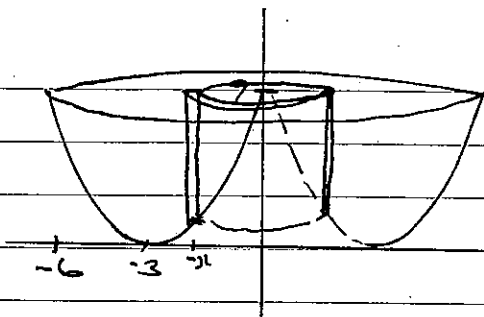
$$= \frac{1}{e}$$

Now Slope of the normal at P

$$= \frac{1}{e} \quad (\text{slope tangent} = -e)$$

$\therefore$ normal at $P \parallel$ to MS'

b)

Volume of a shell = $2\pi rh$

$$= 2\pi xy \Delta x$$

$$= 2\pi x(9-y) \Delta x$$

$$\therefore V = \sum_{x=-6}^0 2\pi x(9-y) \Delta x$$

$$\therefore V = \lim_{\Delta x \rightarrow 0} \sum_{x=-6}^0 2\pi x(9-y) \Delta x$$

$$= 2\pi \int_{-6}^0 x(9-y) dx$$

$$= 2\pi \int_{-6}^0 x(9 - (x+3)^2) dx$$

$$= 2\pi \int_{-6}^0 x(9 - x^2 - 6x - 9) dx$$

$$= 2\pi \int_{-6}^0 -x^3 - 6x^2 dx$$

$$= -2\pi \left[\frac{x^4}{4} + 2x^3 \right]_{-6}^0$$

$$= -2\pi [-108]$$

$$= 216\pi \text{ cubic units}$$

$$c) \frac{dp}{dt} = k\sqrt{p}$$

$$\frac{dp}{dt} = kp^{\frac{1}{2}}$$

$$\frac{dt}{dp} = \frac{1}{kp}^{-\frac{1}{2}}$$

$$t = \frac{2}{k} p^{\frac{1}{2}} + C$$

$$kt = 2\sqrt{p} + C_1$$

$$\text{at } t=0, p=100$$

$$0 = 2\sqrt{100} + C_1$$

$$-20 = C_1$$

$$kt = 2\sqrt{p} - 20$$

$$t=6, p=169$$

$$6k = 2\sqrt{169} - 20$$

$$6k = 26 - 20$$

$$6k = 6$$

$$k=1$$

$$\therefore t = 2\sqrt{p} - 20$$

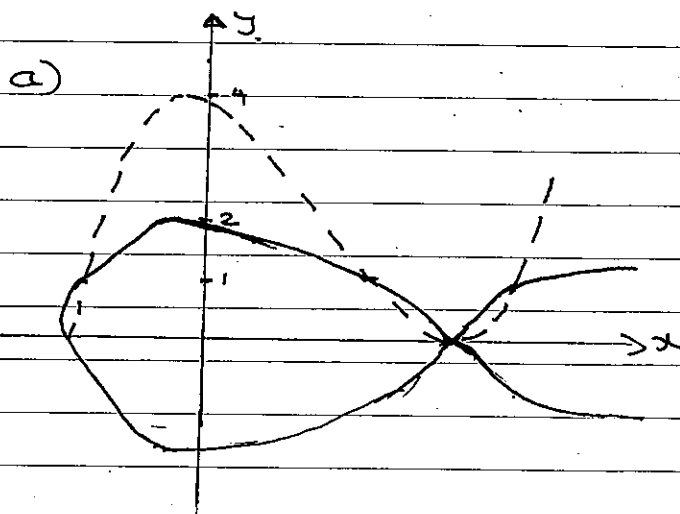
$$\therefore p = \left(\frac{t+20}{2}\right)^2$$

$$\text{at } t=12, p = \left(\frac{12+20}{2}\right)^2$$

$$= 256$$

$\therefore$ after one year there will be 256 fish in the lake.

Question 4



$$b) i) y^2 = x^4(4+x)$$

'x' intercepts: $y=0$

$$x^4(4+x) = 0$$

$$x = 0, -4$$

'y' intercept: $x=0$

$$y^2 = 0$$

$$y = 0$$

$$ii) y^2 = x^4(4+x)$$

$$\text{ie } y^2 = 4x^4 + x^5$$

$$2y \frac{dy}{dx} = 16x^3 + 5x^4$$

$$\frac{dy}{dx} = \frac{16x^3 + 5x^4}{2y}$$

iii) Stationary points: $\frac{dy}{dx} = 0$

$$\frac{5x^4 + 16x^3}{2y} = 0$$

$$\therefore 5x^4 + 16x^3 = 0$$

$$x^3(5x + 16) = 0$$

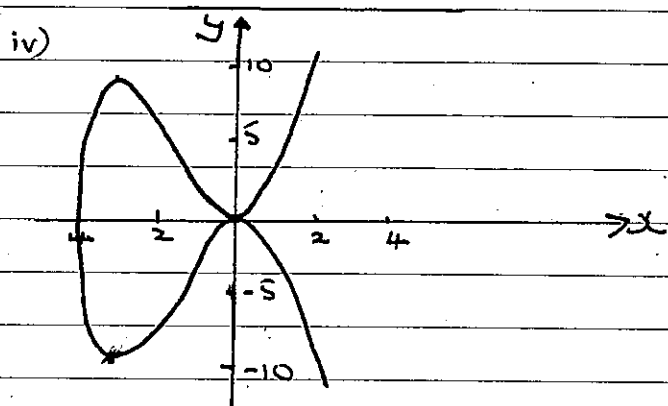
$$x = 0, \quad x = -\frac{16}{5}$$

$$\text{at } x = 0; \quad x = -\frac{16}{5}$$

$$y = 0, \quad y^2 = 83.9$$

$$y = \pm 9.2$$

$\therefore$ S.P. $(0,0), (-3.2, 9.2), (-3.2, -9.2)$



v) $y^2 = x^4(4+x)$

$$A = 2 \int_{-4}^0 y \, dx$$

$$= 2 \int_{-4}^0 x^2 \sqrt{4+x} \, dx$$

Let $u^2 = 4+x$ at $x = -4, u = 0$

$x = u^2 - 4$ at $x = 0, u = 2$

$$\frac{dx}{du} = 2u$$

$$dx = 2u \, du$$

$$= 2 \int_0^2 (u^2 - 4)^2 u \cdot 2 \, du$$

$$= 4 \int_0^2 (u^4 - 8u^2 + 16) u \, du$$

$$= 4 \int_0^2 (u^5 - 8u^3 + 16u) \, du$$

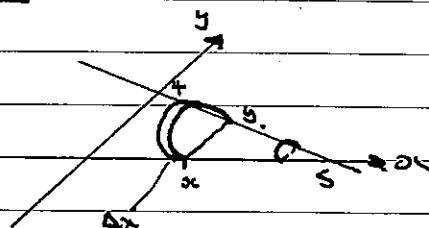
$$= 4 \left[\frac{u^6}{6} - 2u^4 + 8u^2 \right]_0^2$$

$$= 4 \left[10^{2/3} - 0 \right]$$

$$= 4 \cdot 2^{2/3} \text{ Square units}$$

Ques 5

a)



$$\text{Volume of slice} = \frac{\pi}{2} \left(\frac{y}{2} \right)^2 \Delta x$$

$$\therefore V = \sum_{x=0}^5 \frac{\pi}{8} y^2 \Delta x$$

$$V = \lim_{\Delta x \rightarrow 0} \sum_{x=0}^5 \frac{\pi}{8} y^2 \Delta x$$

$$= \frac{\pi}{8} \int_0^5 \left(\frac{20-4x}{5} \right)^2 dx$$

$$= \frac{\pi}{8} \int_0^5 \left(4 - \frac{4}{5}x \right)^2 dx$$

$$= \frac{\pi}{8} \left[\frac{(4 - \frac{4}{5}x)^3}{3 \cdot \frac{4}{5}} \right]_0^5$$

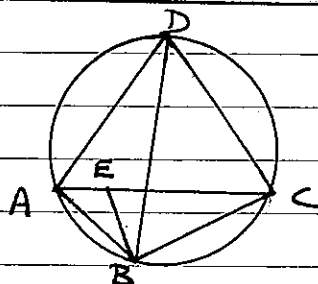
$$= \frac{\pi}{8} \left[\frac{5(4 - \frac{4}{5}x)^3}{-12} \right]_0^5$$

$$= \frac{\pi}{8} \left[0 - \left(\frac{320}{-12} \right) \right]$$

$$= \frac{\pi}{8} \times \frac{320}{12}$$

$$= \frac{10\pi}{3} \text{ cubic units}$$

b)



i) $\angle ABE = \angle DBC$ given
 $\angle BAE = \angle BDC$ angles at the circumference standing on the same arc are equal.

6.

$$\therefore \triangle ABE \parallel \triangle DBC \quad (\text{A.A.A.})$$

$$\text{ii) let } \angle DAC = \alpha$$

$$\therefore \angle DBC = \alpha \quad \text{angle at circumference standing on same arc}$$

$$\therefore \angle ABE = \alpha \quad \text{given}$$

$$\text{let } \angle BAE = \beta$$

$$\therefore \angle DAB = \alpha + \beta$$

$$\text{also } \angle BEC = \alpha + \beta \quad \text{exterior } \angle \text{ of } \triangle ABE$$

$$\therefore \angle DAB = \angle BEC$$

$$\angle ADB = \angle ECB \quad \text{angle at the circumference standing on same arc.}$$

$$\therefore \triangle ABD \parallel \triangle EBC \quad (\text{A.A.A.})$$

$$\text{iii) As } \triangle ABE \parallel \triangle DBC$$

$$\therefore \frac{AB}{BD} = \frac{AE}{DC}$$

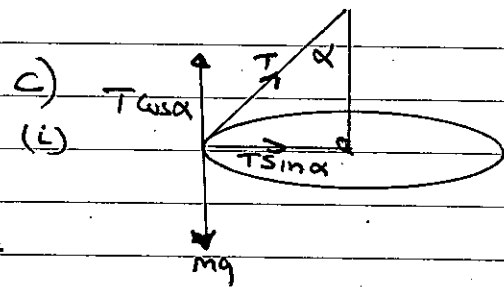
$$\text{ie } AB \times CD = BD \times AE$$

$$\text{Also as } \triangle ABD \parallel \triangle EBC$$

$$\therefore \frac{BC}{DB} = \frac{EC}{AD}$$

Now,

$$\begin{aligned} AB \times CD + BC \times AD &= BD \times AE + BD \times EC \\ &= BD (AE + EC) \\ &= BD \times AC \end{aligned}$$



resolving forces horizontally and vertically

$$T \sin \alpha = \frac{mv^2}{r} \quad \text{--- (1)}$$

$$T \cos \alpha = mg \quad \text{--- (2)}$$

$$\text{from (2) } T = \frac{mg}{\cos \alpha}$$

$$(1) \div (2) \quad \tan \alpha = \frac{v^2}{rg}$$

$$\therefore v^2 = rg \tan \alpha$$

$$v = \sqrt{rg \tan \alpha}$$

ii) If string can sustain a mass of 10 kg $\Rightarrow$ maximum tension in the string = $Mg = 10 \times 10 = 100$ newtons.

Now from (i)

$$\cos \alpha = \frac{mg}{T}$$

$$= \frac{5 \times 10}{100}$$

$$= 0.5$$

$$\therefore \alpha = 60^\circ$$

$$\text{Also } T \sin \alpha = mr\omega^2$$

$$= m(\omega^2 \sin \alpha)$$

$$T = m\omega^2$$

$$\therefore \omega^2 = \frac{T}{mL}$$

$$= \frac{100}{5 \times 0.6}$$

$$= 33.3$$

$$\therefore \omega = 5.77 \text{ rad/sec.}$$

$$= 0.9 \text{ rev/sec}$$

$$= 54 \text{ rev/min.}$$

Ques 6

a) (i) let $P(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4$
with $a_0, a_1, a_2, a_3, a_4 \in \mathbb{R}$.

if α is a zero then $P(\alpha) = 0$

$$\text{i.e. } a_0 + a_1\alpha + a_2\alpha^2 + a_3\alpha^3 + a_4\alpha^4 = 0$$

$$a_0 + a_1\alpha + a_2\alpha^2 + a_3\alpha^3 + a_4\alpha^4 = 0$$

$$a_0 + a_1\bar{\alpha} + a_2\bar{\alpha}^2 + a_3\bar{\alpha}^3 + a_4\bar{\alpha}^4 = 0$$

$$a_0 + a_1\bar{\alpha} + a_2\bar{\alpha}^2 + a_3\bar{\alpha}^3 + a_4\bar{\alpha}^4 = 0$$

$$\therefore P(\bar{\alpha}) = 0$$

ii) $\alpha, \bar{\alpha}$ are zeros of $P(x)$

$\Rightarrow (x - \alpha)(x - \bar{\alpha})$ is a factor
of $P(x)$ ($\alpha \neq \bar{\alpha}$)

$$\begin{aligned} \text{But } (x - \alpha)(x - \bar{\alpha}) &= x^2 - (\alpha + \bar{\alpha})x + \alpha\bar{\alpha} \\ &= x^2 - 2\operatorname{Re}(\alpha)x + |\alpha|^2 \\ &= x^2 - 2ax + r^2 \end{aligned}$$

$\therefore x^2 - 2ax + r^2$ is a factor of $P(x)$

iii) α zero of multiplicity 2

$$\Rightarrow P(\alpha) = P'(\alpha) = 0$$

$$\Rightarrow P(\bar{\alpha}) = P'(\bar{\alpha}) = 0 \quad (\text{from above})$$

$\therefore \bar{\alpha}$ is also a zero of multiplicity 2.

$$\therefore x^4 - 8x^3 + 30x^2 - 56x + 49 = 0$$

has roots $\alpha, \alpha, \bar{\alpha}, \bar{\alpha}$

$$\text{then } 2(\alpha + \bar{\alpha}) = 8 \quad (-b/a)$$

$$\text{i.e. } 4a = 8$$

$$a = 2$$

$$\text{also } (\alpha\bar{\alpha})^2 = 49 \quad (c/a)$$

$$(r^2)^2 = 49$$

$$r^4 = 49$$

$$r = \sqrt{7}$$

$$\Rightarrow \alpha = 2 \pm i\sqrt{7}$$

$$\therefore P(x) = (x^2 - 4x + 7)^2$$

and roots of $P(x) = 0$ are

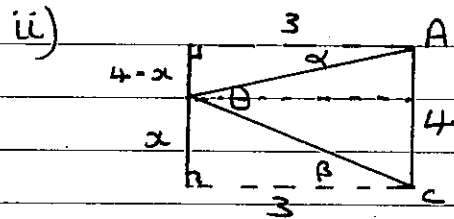
$$2 + i\sqrt{7}, 2 + i\sqrt{7}, 2 - i\sqrt{7}, 2 - i\sqrt{7}$$

$$\text{b) i) } \tan(\alpha + \beta) = \frac{\sin(\alpha + \beta)}{\cos(\alpha + \beta)}$$

$$= \frac{\sin\alpha \cos\beta + \cos\alpha \sin\beta}{\cos\alpha \cos\beta - \sin\alpha \sin\beta}$$

$$= \frac{\frac{\sin\alpha}{\cos\alpha} + \frac{\sin\beta}{\cos\beta}}{1 - \frac{\sin\alpha \sin\beta}{\cos\alpha \cos\beta}} \quad (\div \cos\alpha \cos\beta)$$

$$= \frac{\tan\alpha + \tan\beta}{1 - \tan\alpha \tan\beta}$$



From the diagram:

$$\tan\alpha = \frac{4-x}{3}, \quad \tan\beta = \frac{x}{3}$$

Now $\theta = \alpha + \beta$ (alternate $\angle$'s in parallel lines)

$$\therefore \tan\theta = \tan(\alpha + \beta)$$

$$= \frac{\tan\alpha + \tan\beta}{1 - \tan\alpha \tan\beta}$$

$$= \frac{\frac{4-x}{3} + \frac{x}{3}}{1 - \left(\frac{4-x}{3}\right) \cdot \frac{x}{3}}$$

$$= \frac{\frac{4}{3}}{1 - \frac{x(4-x)}{9}}$$

$$= \frac{12}{9 - x(4-x)}$$

$$= \frac{12}{9 - 4x + x^2}$$

$$\text{iii) } \tan\theta = \frac{12}{9 - 4x + x^2}$$

$$\theta = \tan^{-1}\left(\frac{12}{9 - 4x + x^2}\right)$$

8.

$$\frac{d\theta}{dx} = \frac{1}{1 + \left(\frac{12}{x^2 - 4x + 9}\right)^2} \times \frac{-12(2x-4)}{(x^2 - 4x + 9)^2}$$

$$= \frac{-24(x-2)}{(x^2 - 4x + 9)^2 + 12}$$

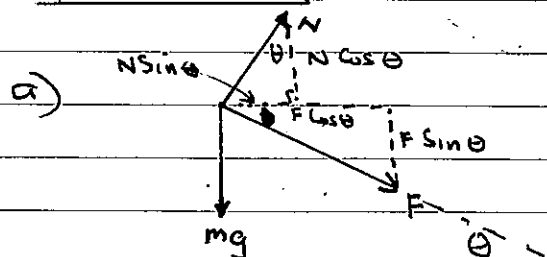
Stationary pt. when $\frac{d\theta}{dx} = 0$
 $-24(x-2) = 0$
 $x = 2.$

$$0 < f'(1), \quad f'(3) < 0$$

$\therefore$ max when $x = 2.$

$\therefore \theta$ max when the billboard is 2m below the eye level.

Question 7



Resolving forces.

Vertically,

$$N \cos \theta - F \sin \theta - mg = 0 \quad \dots (1)$$

$$N \sin \theta + F \cos \theta = \frac{mv^2}{r} \quad \dots (2)$$

$$(1) \times \sin \theta$$

$$N \sin \theta \cos \theta - F \sin^2 \theta = mg \sin \theta \quad \dots (3)$$

$$(2) \times \cos \theta$$

$$N \sin \theta \cos \theta + F \cos^2 \theta = \frac{mv^2 \cos \theta}{r} \quad \dots (4)$$

$$(4) - (3) \quad F(\sin^2 \theta + \cos^2 \theta) = \frac{mv^2 \cos \theta}{r} - mg \sin \theta$$

$$F = \frac{mv^2 \cos \theta}{r} - mg \sin \theta$$

b) i) $xy = c^2$
 $y = \frac{c^2}{x}$

$$\frac{dy}{dx} = -\frac{c^2}{x^2}$$

at $x = ct$: $\frac{dy}{dx} = \frac{-c^2}{c^2 t^2}$
 $= -\frac{1}{t^2}$

$\therefore$ gradient normal $= t^2$

$\therefore$ equation of the normal

$$y - \frac{c}{t} = t^2(x - ct)$$

$$y - \frac{c}{t} = t^2 x - ct^3$$

$$y = t^2 x + \frac{c}{t} - ct^3$$

ii) for N: $y = t^2 x + \frac{c}{t} - ct^3 \quad \dots (1)$

$$y = x \quad \dots (2)$$

equating (1) and (2) $x = t^2 x + \frac{c}{t} - ct^3$

$$tx(1 - t^2) = c - ct^4$$

$$x = \frac{c(1 - t^4)}{t(1 - t^2)}$$

$$= \frac{c(1 + t^2)}{t}$$

$$\therefore N \left(\frac{c(1 + t^2)}{t}, \frac{c(1 + t^2)}{t} \right)$$

For T: $y = -\frac{2c}{t^2} + \frac{2c}{t} \quad \dots (3)$

$$y = x \quad \dots (4)$$

equating (3) & (4) $x = -\frac{2c}{t^2} + \frac{2c}{t}$

$$x(t^2 + 1) = \frac{2ct}{t^2 + 1}$$

$$x = \frac{2ct}{t^2 + 1}$$

$\therefore T \left(\frac{2ct}{t^2 + 1}, \frac{2ct}{t^2 + 1} \right)$

$$OT \cdot ON = \sqrt{\left(\frac{c(1+t^2)}{t}\right)^2 + \left(\frac{c(1+t^2)}{t}\right)^2} \cdot \sqrt{\left(\frac{2ct}{1+t^2}\right)^2 + \left(\frac{2ct}{1+t^2}\right)^2}$$

$$= \sqrt{\frac{2c^2(1+t^2)^2}{t^2}} \cdot \sqrt{\frac{8c^2 t^2}{(1+t^2)^2}}$$

$$= \frac{\sqrt{2} \cdot c(1+t^2)}{t} \times \frac{2\sqrt{2} ct}{1+t^2}$$

$$= 4c^2$$

9.

$$c) (i) \frac{x^2 - x + 1}{x+1} \div \frac{x^3}{x^3 + x^2 - x^2 - x} = \frac{x^2 - x + 1}{x+1} \div \frac{x^2 - x}{x+1} = \frac{x^2 - x + 1}{x^2 - x} = \frac{x^2 - x + 1}{x(x-1)}$$

$$\therefore x^3 = (x+1)(x^2 - x + 1) - 1$$

$$\text{OR } \frac{x^3}{x+1} = x^2 - x + 1 - \frac{1}{x+1}$$

$$ii) \int \frac{dx}{x^{1/2} + x^{1/3}}$$

$$\text{Let } x = u^6$$

$$\frac{dx}{du} = 6u^5$$

$$dx = 6u^5 du$$

$$\text{also } x^{1/2} = u^3, \quad x^{1/3} = u^2$$

$$= \int \frac{6u^5}{u^3 + u^2} du$$

$$= \int \frac{6u^3}{u+1} du$$

$$= 6 \int \frac{u^3}{u+1} du$$

$$= 6 \int u^2 - u + 1 - \frac{1}{u+1} du$$

(from ii) above)

$$= 6 \left[\frac{u^3}{3} - \frac{u^2}{2} + u - \ln(u+1) \right]$$

$$= 2u^3 - 3u^2 + 6u - 6\ln(u+1) + C$$

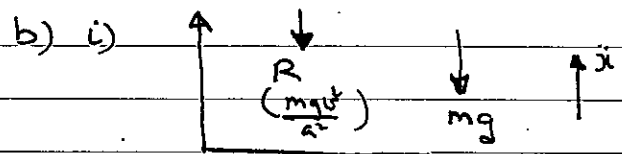
$$= 2x^{1/2} - 3x^{1/3} + 6x^{1/6} - 6\ln(x^{1/6} + 1) + C$$

Ques 8

$$a) \ddot{x} = \frac{dv}{dt}$$

$$= \frac{dv}{dx} \cdot \frac{dx}{dt}$$

$$= v \frac{dv}{dx}$$



$$m\ddot{x} = -mg - R$$

$$m\ddot{x} = -mg - \frac{mgv^2}{a^2}$$

$$\ddot{x} = -g - \frac{g}{a^2} v^2$$

$$v \frac{dv}{dx} = -g \left(1 + \frac{v^2}{a^2} \right)$$

$$= -g \left(\frac{a^2 + v^2}{a^2} \right)$$

$$= -\frac{g}{a^2} (a^2 + v^2)$$

$$ii) v \frac{dv}{dx} = -\frac{g}{a^2} (a^2 + v^2)$$

$$\frac{dv}{dx} = -\frac{g}{a^2} \left(\frac{a^2 + v^2}{v} \right)$$

$$\frac{dx}{dv} = -\frac{a^2}{g} \left(\frac{v}{a^2 + v^2} \right)$$

$$x = -\frac{a^2}{2g} \ln(a^2 + v^2) + C$$

$$\text{at } t=0, x=0, v=u$$

$$0 = -\frac{a^2}{2g} \ln(a^2 + u^2)$$

$$C = \frac{a^2}{2g} \ln(a^2 + u^2)$$

$$x = -\frac{a^2}{2g} \ln(a^2 + v^2) + \frac{a^2}{2g} \ln(a^2 + u^2)$$

$$x = \frac{a^2}{2g} \ln \left(\frac{a^2 + u^2}{a^2 + v^2} \right)$$

for greatest height $v = 0$

$$x = \frac{a^2}{2g} \ln \left(\frac{a^2 + u^2}{a^2} \right)$$

$$= \frac{a^2}{2g} \ln \left(1 + \frac{u^2}{a^2} \right)$$

c) i) $P(x) = x^n + ax - b$

$$P'(x) = nx^{n-1} + a$$

if x is a double root

$$P(x) = P'(x) = 0$$

$$P'(x) = nx^{n-1} + a$$

$$\therefore nx^{n-1} + a = 0$$

$$x^{n-1} = -\frac{a}{n}$$

ii) $P(x) = 0$

$$x^n + ax - b = 0$$

$$\text{but } a = -nx^{n-1}$$

$$x^n + (-nx^{n-1})x - b = 0$$

$$x^n - nx^n - b = 0$$

$$x^n(1-n) = b$$

$$x^n = \frac{b}{1-n}$$

Now $(x^n)^{n-1} = (x^{n-1})^n$

$$\therefore \left(\frac{b}{1-n} \right)^{n-1} = \left(-\frac{a}{n} \right)^n$$

$$(-1)^{n-1} \left(\frac{b}{1-n} \right)^{n-1} = (-1)^n \left(\frac{a}{n} \right)^n$$

$$(-1)^n \left(\frac{a}{n} \right)^n - (-1)^{n-1} \left(\frac{b}{1-n} \right)^{n-1} = 0$$

$$(-1)^n \left(\frac{a}{n} \right)^n + (-1)^n \left(\frac{b}{1-n} \right)^{n-1} = 0$$

$$\left(\frac{a}{n} \right)^n + \left(\frac{b}{1-n} \right)^{n-1} = 0$$

iii)
$$x = \frac{x^n}{x^{n-1}}$$

$$= \frac{\frac{b}{1-n}}{-\frac{a}{n}}$$

$$= \frac{b}{1-n} \times \frac{n}{-a}$$

$$= \frac{nb}{a(n-1)}$$